

IM1 Book 1 Selected Answers

IM1 Dream Team

September 15, 2026

1. (a) $A_{ABCD} = 25$, $A_{BCEF} = 9$
(b) –
(c) –
(d) $A = 34$
(e) $l = \sqrt{34}$
(f) –
2. $l = 4\sqrt{5}$
3. –
4. $AB = \sqrt{41}$
5. $l = 5\sqrt{2}$
6. $l = \sqrt{5}$, No
7. $d = 10\sqrt{2}$
8. 12
9. $(12, 2), (2, 2)$
10. No
11. (a)

$$\begin{aligned}SR &= PQ = 13 \\RQ &= PS = 17 \\m_{SR} &= m_{PQ} = \frac{-5}{12} \\m_{RQ} &= m_{PS} = \frac{15}{8}\end{aligned}$$

- (b) It is a parallelogram but not a rectangle.
12. $(-6, 3), (10, 3)$

13. $AB = BC = \sqrt{10}$
14. (a) $(5, 0)$ – Answers may vary.
 (b) $(5, 1)$ – Answers may vary.
 (c) $x = 5$
 (d) –
15. (a) $AP = BP = 2\sqrt{5}$
 (b) $(2, 2), (3, 5), (4, 8)$
 (c) No
 (d) $y = 3(x - 1) - 1$
16. (a) $(2, 5), (3, 4)$ – Answers may vary.
 (b) $(1, 4), (2, 3)$ – Answers may vary.
17. $(1, 5)$ – Answers may vary.
 There are infinite options.
18. (a) Same size, different shape, incomparable orientations – the pieces aren't equal.
 (b) Same size, same shape, different orientations – the pieces are equal
- Definition: Two figures are *congruent* if they have the same size and shape.
19. $y = \frac{1}{3}(x - 2) + 2$